10.9.5 Wrap-Up: Growth Mindset

Complete the following sentences based on today's learning target:

I can use proportional relationships to make predictions about a population.

1. A barrier to my learning today was …

2. I will try to overcome this by …

Unit 10, Lesson 10: Making Population Predictions – Part 2

Benchmark

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

Learning Target

☐ I can use proportional relationships to make predictions about a population.

10.10.1 Warm-Up: Bell Work

Antonia and her aunt go fishing at Lake Okeechobee every Sunday. For the past four Sundays, Antonia has kept track of what kinds of fish they caught.

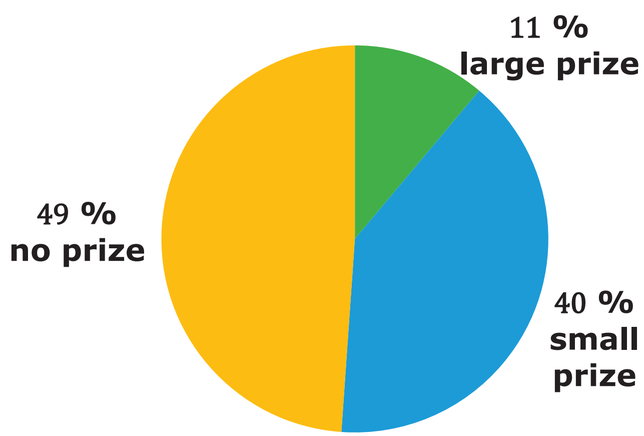
	Bluegill	Bream	Crappie	Largemouth Bass
Number of Fish Caught Last Month	10	12	20	8
Expected Number of Fish to Be Caught Next Sunday				

1. Complete the table to show what type and how many fish Antonia and her aunt are likely to catch next Sunday when they go fishing.

10.10.2 Guided Practice: Making Population Predictions

1. Ella gathered data from a Rubber Duck Pool Game. The circle graph below shows the percentage of people who won a large prize, a small prize, and no prize.

Prizes Won in Rubber Duck Game



There are 180 rubber ducks floating in the pool. The person playing the game takes out one duck and looks for a mark on the bottom of it.

- If there is a blue mark, the person wins a small prize.
- If there is a green mark, the person wins a large prize.
- If there is no mark, the person does not win a prize.

Ella uses her data to predict how many of each kind of duck are in the pool using three different strategies. Her work is shown.

Method 1	Method 2	Method 3
40% of all ducks $\frac{40}{100} = \frac{4}{10} = \frac{72}{180}$ So, 72 ducks have a blue mark.	11% of all ducks $0.11 \times 180 = 19.8$ So, 20 ducks have a green mark.	49% of all ducks $\frac{49}{100} \times \frac{0.56}{0.56} = \frac{27.44}{180}$ So, 27 ducks have no mark.

a. Ella's solution ○ is ○ is not correct because

- ○ the number of ducks with green marks is less than the number of ducks with blue or no marks.
- ○ the number of ducks with no marks is less than the number of ducks with blue marks.
- ○ Ella's solution shows a total of 180 ducks.

b. Write an explanation for why you agree or disagree with Ella's solution. Use specific examples from Ella's strategies to justify your opinion.

c. Compare your explanation with your partner's. Discuss any differences and make changes if necessary.

d. Ella used three different solution strategies. In the bottom of each column in Ella's table, label each strategy with the correct name from the descriptions.

When using a **scale factor,** first divide the population by the sample to find the scale factor, then multiply the numerator and denominator of the sample fraction by the scale factor.

When using an **equivalent fraction,** write multiple fractions equal to the sample fraction until the denominator shows the total population.

When using a **percent,** multiply the total population by the decimal or fraction form of the percent of the sample.

2. Bianca picks blocks out of a bag one at a time. She picks out a block, records the color, and puts the block back in the bag. She cannot see the blocks, but she knows there are 6 blocks in the bag. Of the 50 times she picked a block out of the bag, she picked a blue block 29 times, a red block 10 times, and a yellow block 11 times.

   a. Write ratios to show the number of times Bianca picked a blue, red, and yellow block out of the bag.

   Blue ______ Red ______ Yellow _______

b. Explain which strategy will work best to predict how many of the 6 blocks are blue, red, and yellow.

c. Using the strategy you selected, how many of the 6 blocks are blue, red, and yellow? Show your work.

d. Explain whether the strategy you chose was the best strategy to use, or if another would have worked better.

10.10.3 Collaboration: Stocking a Vending Machine

Students at Sanford Middle School are upset the school vending machine is always sold out of their favorite items. The student council decides to survey students to find a better way to stock the vending machine. They randomly sample 60 students. Each student surveyed picked their 5 favorite items from the vending machine. The results are shown in the table.

Item	Number of Times Chosen in Survey	Number that Should be Stocked
Reese’s	30	
Skittles	24	
Starburst	21	
Snickers	42	
Pretzels	24	
Chips	30	
Takis	39	
Gum	36	
M&Ms	30	
Trail Mix	24	

The vending machine at school has 30 different snack compartments, each with 15 metal spirals for each snack. The vending machine can hold 450 total snacks.

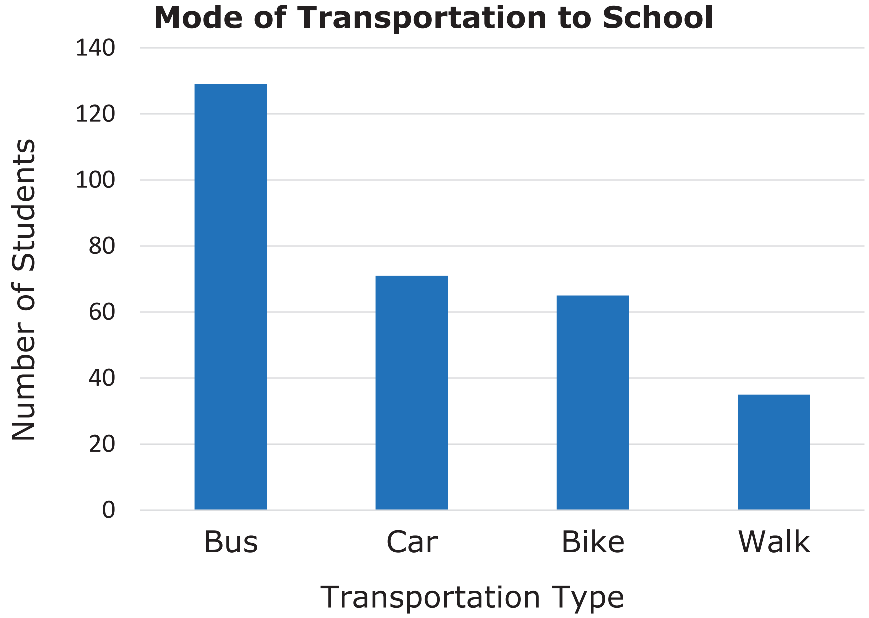
1. Work with your partner to decide how many of each snack should be stocked in the vending machine. Complete the table with your decision.

   Your work will be assessed using these criteria.

   - Each of the strategies: scale factor, equivalent fraction, and percent are used. If one of the strategies is not used, there is an explanation for why the strategy was not used.
   - It is clear, either from your work or a written explanation, why you have stocked the vending machine the way you did.
   - The number of each snack stocked in the vending machine is a direct reflection of the survey results.

10.10.4 Your Turn!

1. Suwannee County School District is studying changes that will need to be made to the transportation department when building new schools. To determine this, it is important to understand how students get to school. The school district has a population of $5{,}959$ students, so a random sample of approximately 5% of the students in each school was conducted. The results of the survey are shown in the bar chart.



   a. Show how to predict the number of students in the district who take the bus to get to school.

b. Show how to predict the number of students in the district who ride their bike to school.

c. Whether or not a student can take the bus to school depends on how close they live to the school. Explain how this data set does not take this into consideration and what effect that might have on the district's decision.

10.10.5 Wrap-Up: Ticket Out the Door

Lin has one month before the seventh-grade class presidential election, and she wants to determine how likely she is to win. Since she can't ask all 665 seventh graders who they are voting for, she conducts a random sample of 105 seventh-grade students. The results of the survey are in the table.

Candidate	Number of People Planning to Vote for the Candidate
Lin	42
Beslinda (Lin's opponent)	30
Undecided	33

1. Use either percent, scale factor, or equivalent fraction to predict how many of the 665 total seventh-grade students currently plan on voting for Lin. Show your work.

2. Use another strategy to predict how many of the total seventh-grade voters are still undecided.